

Forced Vibrations

*Mechanical Spring–Mass–Damper Systems:
Resonance, Amplitude Peaking & Beats*

Prepared by Staples Education

Spring–Mass–Damper Model • Periodic Driving Forces • Undamped Response • Beats •
Pure Resonance • Transient & Steady State • Practical Resonance • Amplitude Peaking

Contents

1	Part I Condensed Cheat Sheet	2
1.1	The Spring–Mass–Damper Model	2
1.2	Undamped Forcing: Non-Resonant Response	2
1.3	Beats	3
1.4	Pure Resonance (Undamped)	3
1.5	Damped Forced Vibrations: Transient and Steady State	4
1.6	Practical Resonance and Amplitude Peaking	5
2	Part II Review Guide	6
3	Part III Answer Key & Worked Solutions	7

1 Part I | Condensed Cheat Sheet

1.1 The Spring–Mass–Damper Model

A mass m on a spring of stiffness k , with a damper of coefficient γ and an external driving force $F(t)$, obeys Newton's second law in the form below. Every result in this guide is a consequence of this single equation — the behavior changes only with the size of γ and how the driving frequency ω compares to the system's natural frequency.

Equation of motion

$$m u'' + \gamma u' + k u = F(t), \quad \gamma \geq 0, \ k > 0.$$

Natural angular frequency (undamped, unforced):

$$\omega_0 = \sqrt{\frac{k}{m}}.$$

For periodic forcing we take $F(t) = F_0 \cos(\omega t)$, where ω is the *driving frequency* and F_0 the forcing amplitude.

The complete solution always splits into two pieces — the *transient* (the homogeneous solution, governed by m, γ, k) and the *steady state* (a particular solution tracking $F(t)$). When $\gamma > 0$ the transient decays and only the steady state survives; when $\gamma = 0$ neither piece decays, which is exactly what makes undamped resonance and beats possible.

1.2 Undamped Forcing: Non-Resonant Response

Set $\gamma = 0$ and assume $\omega \neq \omega_0$. The trial solution $U = A \cos \omega t$ contains no derivative-of-cosine term, so a single coefficient is determined directly.

Non-resonant steady state ($\omega \neq \omega_0$):

$$A = \frac{F_0}{k - m\omega^2} = \frac{F_0}{m(\omega_0^2 - \omega^2)}, \quad U(t) = A \cos \omega t.$$

General solution:

$$u(t) = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t + \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos \omega t.$$

Worked Example — non-resonant steady state. Find the steady-state response of $u'' + 4u = 2 \cos 3t$.

Step 1. Read off parameters: $m = 1$, $k = 4$, $\omega_0 = 2$, $\omega = 3$, $F_0 = 2$.

Step 2. Apply the amplitude formula:

$$A = \frac{F_0}{k - m\omega^2} = \frac{2}{4 - 9} = -\frac{2}{5}.$$

Step 3. State the steady state:

$$U(t) = -\frac{2}{5} \cos 3t.$$

The negative coefficient means the response is 180° out of phase with the drive — typical when forcing above the natural frequency.

1.3 Beats

When ω is *close* to ω_0 (but $\gamma = 0$ and $\omega \neq \omega_0$), starting from rest produces a striking pattern: a rapid oscillation whose amplitude swells and fades. Begin from $u(0) = 0$, $u'(0) = 0$. The solution collapses to a difference of cosines, which a product-to-sum identity rewrites as an envelope times a carrier.

Beats (from rest, $\omega \neq \omega_0$, $\gamma = 0$):

$$u(t) = \frac{F_0}{m(\omega_0^2 - \omega^2)} (\cos \omega t - \cos \omega_0 t).$$

Using $\cos \omega t - \cos \omega_0 t = 2 \sin\left(\frac{\omega_0 - \omega}{2} t\right) \sin\left(\frac{\omega_0 + \omega}{2} t\right)$:

$$u(t) = \underbrace{\frac{2F_0}{m(\omega_0^2 - \omega^2)} \sin\left(\frac{\omega_0 - \omega}{2} t\right)}_{\text{slow envelope}} \underbrace{\sin\left(\frac{\omega_0 + \omega}{2} t\right)}_{\text{fast carrier}}.$$

Beat (amplitude) angular frequency: $|\omega_0 - \omega|$, **beat period:** $\frac{2\pi}{|\omega_0 - \omega|}$.

Worked Example — beats. Solve $u'' + 100u = 3 \cos 8t$ with $u(0) = 0$, $u'(0) = 0$.

Step 1. Here $\omega_0 = 10$, $\omega = 8$, $F_0 = 3$, so the non-resonant coefficient is

$$A = \frac{3}{100 - 64} = \frac{3}{36} = \frac{1}{12}.$$

Step 2. Impose the rest conditions on $u = c_1 \cos 10t + c_2 \sin 10t + \frac{1}{12} \cos 8t$:

$$u(0) = c_1 + \frac{1}{12} = 0 \Rightarrow c_1 = -\frac{1}{12}, \quad u'(0) = 10c_2 = 0 \Rightarrow c_2 = 0.$$

Step 3. Combine and apply the identity:

$$u = \frac{1}{12} (\cos 8t - \cos 10t) = \frac{1}{12} \cdot 2 \sin t \sin 9t = \frac{1}{6} \sin t \sin 9t.$$

Step 4. Identify the structure: envelope $\frac{1}{6} \sin t$ (slow), carrier $\sin 9t$ (fast); beat period $2\pi/|10 - 8| = \pi$.

1.4 Pure Resonance (Undamped)

If $\omega = \omega_0$ exactly and $\gamma = 0$, the trial $\cos \omega_0 t$ already solves the homogeneous equation, so undetermined coefficients requires an extra factor of t . The amplitude then grows *without bound*.

Pure resonance ($\omega = \omega_0$, $\gamma = 0$): the particular solution is

$$U(t) = \frac{F_0}{2m\omega_0} t \sin \omega_0 t.$$

The factor t forces the oscillation amplitude $\frac{F_0}{2m\omega_0} t$ to increase linearly forever.

Worked Example — pure resonance. Solve $u'' + 9u = 6 \cos 3t$.

Step 1. Note $\omega_0 = 3 = \omega$ — this is resonance.

Step 2. Use the resonance formula with $m = 1$, $F_0 = 6$:

$$U(t) = \frac{F_0}{2m\omega_0} t \sin \omega_0 t = \frac{6}{2(1)(3)} t \sin 3t = t \sin 3t.$$

Step 3. Verify the growth: the envelope $\pm t$ expands linearly, so $|u|$ is unbounded as $t \rightarrow \infty$.

Theorem (Undamped Resonance). For $mu'' + ku = F_0 \cos(\omega t)$ with $\omega_0 = \sqrt{k/m}$, the response is bounded for all t when $\omega \neq \omega_0$, but grows without bound (amplitude $\propto t$) when $\omega = \omega_0$. Any positive damping removes this unbounded growth.

1.5 Damped Forced Vibrations: Transient and Steady State

With $\gamma > 0$ the homogeneous solution decays — it is the transient. What remains is a sinusoid at the driving frequency, lagging the force by a phase δ .

Steady-state response for $mu'' + \gamma u' + ku = F_0 \cos \omega t$:

$$U(t) = R \cos(\omega t - \delta),$$

$$R = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + \gamma^2 \omega^2}} = \frac{F_0}{\sqrt{m^2(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2}},$$

$$\tan \delta = \frac{\gamma \omega}{k - m\omega^2}, \quad 0 \leq \delta \leq \pi.$$

(If $k - m\omega^2 < 0$, take δ in the second quadrant, i.e. $\delta > \pi/2$.)

Worked Example — damped steady state. Find the steady-state response of $u'' + 3u' + 2u = 10 \cos t$, its amplitude, and phase.

Step 1. Parameters: $m = 1$, $\gamma = 3$, $k = 2$, $\omega = 1$, $F_0 = 10$; auxiliary quantity $k - m\omega^2 = 2 - 1 = 1$.

Step 2. Amplitude:

$$R = \frac{10}{\sqrt{(1)^2 + (3)^2(1)^2}} = \frac{10}{\sqrt{10}} = \sqrt{10}.$$

Step 3. Phase:

$$\tan \delta = \frac{\gamma \omega}{k - m\omega^2} = \frac{3}{1} = 3 \Rightarrow \delta = \arctan 3 \approx 1.249 \text{ rad.}$$

Step 4. Equivalent rectangular form ($A \cos t + B \sin t$):

$$A = \frac{F_0(k - m\omega^2)}{(k - m\omega^2)^2 + \gamma^2\omega^2} = \frac{10}{10} = 1, \quad B = \frac{F_0\gamma\omega}{10} = \frac{30}{10} = 3,$$

$$U(t) = \cos t + 3 \sin t = \sqrt{10} \cos(t - \delta).$$

1.6 Practical Resonance and Amplitude Peaking

With damping, no frequency makes the amplitude infinite — but $R(\omega)$ may still *peak*. Minimizing the denominator $m^2(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2$ in R locates that peak.

Practical resonance frequency (amplitude peak):

$$\omega_{\max} = \sqrt{\omega_0^2 - \frac{\gamma^2}{2m^2}}, \quad \text{exists only if } \gamma^2 < 2mk.$$

Peak amplitude:

$$R_{\max} = \frac{F_0}{\gamma \sqrt{\omega_0^2 - \frac{\gamma^2}{4m^2}}} = \frac{F_0}{\gamma \omega_0 \sqrt{1 - \frac{\gamma^2}{4mk}}} \approx \frac{F_0}{\gamma \omega_0} \quad (\text{light damping}).$$

Two consequences worth memorizing — the peak sits *below* ω_0 and slides lower as γ grows; and once $\gamma^2 \geq 2mk$ the peak disappears entirely, so R decreases monotonically from its static value F_0/k .

Worked Example — amplitude peaking. For $u'' + 2u' + 5u = 4 \cos \omega t$, find $\omega_{\max}$ and $R_{\max}$.

Step 1. Parameters: $m = 1$, $\gamma = 2$, $k = 5$, $F_0 = 4$, $\omega_0^2 = 5$.

Step 2. Check existence: $\gamma^2 = 4 < 2mk = 10$ — a peak exists.

Step 3. Peak frequency:

$$\omega_{\max} = \sqrt{\omega_0^2 - \frac{\gamma^2}{2m^2}} = \sqrt{5 - \frac{4}{2}} = \sqrt{3}.$$

Step 4. Peak amplitude:

$$R_{\max} = \frac{F_0}{\gamma \sqrt{\omega_0^2 - \gamma^2/(4m^2)}} = \frac{4}{2\sqrt{5-1}} = \frac{4}{2 \cdot 2} = 1.$$

Theorem (Existence of Practical Resonance). The steady-state amplitude $R(\omega)$ of a damped, periodically forced oscillator attains an interior maximum at $\omega_{\max} = \sqrt{\omega_0^2 - \gamma^2/(2m^2)}$ if and only if $\gamma^2 < 2mk$. Otherwise $R(\omega)$ is strictly decreasing on $\omega \geq 0$ and its largest value is the static deflection F_0/k .

2 Part II | Review Guide

Ten problems, ordered from basic setup to full synthesis. Solutions follow in Part III.

1. A 2 kg mass is attached to a spring of stiffness $k = 18$ N/m. Neglecting damping, it is driven by $F(t) = 6 \cos(2t)$ N. Write the equation of motion and find the natural angular frequency ω_0 .
2. Find the steady-state (particular) solution of $u'' + 4u = 3 \cos t$.
3. Consider $u'' + 25u = F_0 \cos(\omega t)$. For which driving frequency ω does pure (undamped) resonance occur, and what is the form of the resulting particular solution?
4. Solve the IVP $u'' + 16u = 8 \cos(4t)$, $u(0) = 0$, $u'(0) = 0$, and describe the long-term behavior.
5. Solve $u'' + 81u = \cos(8t)$ with $u(0) = 0$, $u'(0) = 0$. Rewrite the solution as an envelope times a rapid oscillation, and identify the beat period.
6. A damped system satisfies $u'' + 2u' + 5u = 4 \cos t$. Find the steady-state response, its amplitude R , and its phase lag δ .
7. For $u'' + 2u' + 10u = 6 \cos(\omega t)$ the driving frequency is tunable. Find the frequency $\omega_{\max}$ giving the maximum steady-state amplitude (practical resonance) and the corresponding maximum amplitude $R_{\max}$.
8. A system has $m = 2$ kg and $k = 8$ N/m. (a) Determine the range of damping coefficients γ for which practical resonance occurs. (b) Compare this threshold with the critical-damping value $\gamma_{\text{cr}} = 2\sqrt{mk}$ and interpret.
9. Solve the full IVP $u'' + 2u' + 2u = 2 \cos t$, $u(0) = 0$, $u'(0) = 0$, separating the transient and steady-state parts.
10. A 1 kg mass hangs from a spring with $k = 25$ N/m and damping $\gamma = 3$ N·s/m, driven by $F(t) = 10 \cos(\omega t)$ N. (a) Find ω_0 . (b) Does practical resonance occur? (c) Find $\omega_{\max}$. (d) Compare the steady-state amplitude at $\omega_{\max}$ with the amplitude at $\omega = \omega_0$ and interpret.

3 Part III | Answer Key & Worked Solutions

Solution 1

The model $mu'' + ku = F(t)$ with $m = 2$, $k = 18$ gives

$$2u'' + 18u = 6 \cos 2t, \quad \text{or} \quad u'' + 9u = 3 \cos 2t.$$

The natural frequency is

$$\omega_0 = \sqrt{k/m} = \sqrt{18/2} = \sqrt{9} = 3 \text{ rad/s.}$$

Since the drive frequency $\omega = 2 \neq 3 = \omega_0$, the response is non-resonant.

Solution 2

Here $m = 1$, $k = 4$, $\omega_0 = 2$, $\omega = 1$, $F_0 = 3$. Trial $U = A \cos t$:

$$A = \frac{F_0}{k - m\omega^2} = \frac{3}{4 - 1} = 1.$$

$$U(t) = \cos t.$$

Solution 3

The natural frequency is $\omega_0 = \sqrt{25} = 5$, so pure resonance occurs when

$$\omega = \omega_0 = 5.$$

With a repeated forcing frequency, the particular solution carries a factor of t :

$$U(t) = \frac{F_0}{2m\omega_0} t \sin \omega_0 t = \frac{F_0}{2(1)(5)} t \sin 5t = \frac{F_0}{10} t \sin 5t.$$

Solution 4

Resonance: $\omega_0 = 4 = \omega$, $m = 1$, $F_0 = 8$. Particular solution:

$$U(t) = \frac{F_0}{2m\omega_0} t \sin \omega_0 t = \frac{8}{2(1)(4)} t \sin 4t = t \sin 4t.$$

General solution and rest conditions:

$$u = c_1 \cos 4t + c_2 \sin 4t + t \sin 4t,$$

$$u(0) = c_1 = 0, \quad u'(0) = 4c_2 + 0 = 0 \Rightarrow c_2 = 0.$$

$$u(t) = t \sin 4t.$$

Long-term behavior: the amplitude envelope $\pm t$ grows linearly, so $|u| \rightarrow \infty$ as $t \rightarrow \infty$ — unbounded resonance.

Solution 5

Parameters $\omega_0 = 9$, $\omega = 8$, $F_0 = 1$; non-resonant coefficient:

$$A = \frac{1}{81 - 64} = \frac{1}{17}.$$

Apply rest conditions to $u = c_1 \cos 9t + c_2 \sin 9t + \frac{1}{17} \cos 8t$:

$$u(0) = c_1 + \frac{1}{17} = 0 \Rightarrow c_1 = -\frac{1}{17}, \quad u'(0) = 9c_2 = 0 \Rightarrow c_2 = 0.$$

$$u = \frac{1}{17} (\cos 8t - \cos 9t).$$

Product-to-sum, with $\frac{\omega_0 - \omega}{2} = \frac{1}{2}$ and $\frac{\omega_0 + \omega}{2} = \frac{17}{2}$:

$$u(t) = \frac{2}{17} \sin\left(\frac{t}{2}\right) \sin\left(\frac{17t}{2}\right).$$

The slow envelope is $\frac{2}{17} \sin(t/2)$ and the carrier is $\sin(17t/2)$. The beat (amplitude) angular frequency is $|\omega_0 - \omega| = 1$, so the

$$\text{beat period} = \frac{2\pi}{|\omega_0 - \omega|} = 2\pi.$$

Solution 6

Parameters $m = 1$, $\gamma = 2$, $k = 5$, $\omega = 1$, $F_0 = 4$; $k - m\omega^2 = 5 - 1 = 4$.

$$R = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + \gamma^2\omega^2}} = \frac{4}{\sqrt{4^2 + 2^2}} = \frac{4}{\sqrt{20}} = \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5}.$$

$$\tan \delta = \frac{\gamma\omega}{k - m\omega^2} = \frac{2}{4} = \frac{1}{2} \Rightarrow \delta = \arctan \frac{1}{2} \approx 0.4636 \text{ rad}.$$

Rectangular form:

$$A = \frac{F_0(k - m\omega^2)}{20} = \frac{16}{20} = \frac{4}{5}, \quad B = \frac{F_0\gamma\omega}{20} = \frac{8}{20} = \frac{2}{5},$$

$$U(t) = \frac{4}{5} \cos t + \frac{2}{5} \sin t = \frac{2\sqrt{5}}{5} \cos(t - 0.4636).$$

Solution 7

Parameters $m = 1$, $\gamma = 2$, $k = 10$, $F_0 = 6$, $\omega_0^2 = 10$. Existence check: $\gamma^2 = 4 < 2mk = 20$ — a peak exists.

$$\omega_{\max} = \sqrt{\omega_0^2 - \frac{\gamma^2}{2m^2}} = \sqrt{10 - 2} = \sqrt{8} = 2\sqrt{2} \text{ rad/s}.$$

$$R_{\max} = \frac{F_0}{\gamma\sqrt{\omega_0^2 - \gamma^2/(4m^2)}} = \frac{6}{2\sqrt{10 - 1}} = \frac{6}{2 \cdot 3} = 1.$$

Solution 8

(a) Practical resonance requires $\gamma^2 < 2mk$. With $m = 2$, $k = 8$:

$$\gamma^2 < 2(2)(8) = 32 \Rightarrow 0 < \gamma < \sqrt{32} = 4\sqrt{2} \approx 5.66 \text{ N}\cdot\text{s/m}.$$

(b) Critical damping:

$$\gamma_{\text{cr}} = 2\sqrt{mk} = 2\sqrt{16} = 8 \text{ N}\cdot\text{s/m}.$$

Since $4\sqrt{2} \approx 5.66 < 8 = \gamma_{\text{cr}}$, the amplitude peak vanishes *while the system is still underdamped*. Interpretation — a system can oscillate freely (underdamped transient) yet show *no* resonant peak in its forced response. Practical resonance is the stricter condition.

Solution 9

Parameters $m = 1$, $\gamma = 2$, $k = 2$, $\omega = 1$, $F_0 = 2$.

Transient. Characteristic equation:

$$r^2 + 2r + 2 = 0 \Rightarrow r = -1 \pm i, \quad u_c = e^{-t}(c_1 \cos t + c_2 \sin t).$$

Steady state. With $k - m\omega^2 = 1$:

$$A = \frac{F_0(k - m\omega^2)}{(k - m\omega^2)^2 + \gamma^2\omega^2} = \frac{2(1)}{1 + 4} = \frac{2}{5}, \quad B = \frac{F_0\gamma\omega}{5} = \frac{4}{5},$$
$$u_p = \frac{2}{5} \cos t + \frac{4}{5} \sin t.$$

Apply initial conditions to $u = u_c + u_p$:

$$u(0) = c_1 + \frac{2}{5} = 0 \Rightarrow c_1 = -\frac{2}{5}.$$
$$u' = e^{-t}[(-c_1 + c_2) \cos t + (-c_1 - c_2) \sin t] - \frac{2}{5} \sin t + \frac{4}{5} \cos t,$$
$$u'(0) = (-c_1 + c_2) + \frac{4}{5} = 0 \Rightarrow \frac{2}{5} + c_2 + \frac{4}{5} = 0 \Rightarrow c_2 = -\frac{6}{5}.$$

Complete solution:

$$u(t) = \underbrace{e^{-t}\left(-\frac{2}{5} \cos t - \frac{6}{5} \sin t\right)}_{\text{transient}} + \underbrace{\frac{2}{5} \cos t + \frac{4}{5} \sin t}_{\text{steady state}}.$$

The transient decays like e^{-t} ; the long-run amplitude is $\sqrt{(2/5)^2 + (4/5)^2} = \frac{2\sqrt{5}}{5}$.

Solution 10

Parameters $m = 1$, $k = 25$, $\gamma = 3$, $F_0 = 10$.

(a) Natural frequency:

$$\omega_0 = \sqrt{k/m} = \sqrt{25} = 5 \text{ rad/s}.$$

(b) Existence of practical resonance:

$$\gamma^2 = 9 < 2mk = 50 \quad \checkmark \text{ — yes, a peak exists.}$$

(c) Peak frequency:

$$\omega_{\max} = \sqrt{\omega_0^2 - \frac{\gamma^2}{2m^2}} = \sqrt{25 - \frac{9}{2}} = \sqrt{20.5} \approx 4.528 \text{ rad/s}.$$

(d) Peak amplitude versus amplitude at $\omega = \omega_0$:

$$R_{\max} = \frac{F_0}{\gamma\sqrt{\omega_0^2 - \gamma^2/(4m^2)}} = \frac{10}{3\sqrt{25 - 2.25}} = \frac{10}{3\sqrt{22.75}} \approx 0.699 \text{ m},$$
$$R(\omega_0) = \frac{F_0}{\sqrt{m^2(\omega_0^2 - \omega_0^2)^2 + \gamma^2\omega_0^2}} = \frac{10}{\sqrt{\gamma^2\omega_0^2}} = \frac{10}{3 \cdot 5} = \frac{10}{15} \approx 0.667 \text{ m}.$$

Interpretation: the true peak (≈ 0.699 m) sits just below ω_0 at $\omega_{\max} \approx 4.53$ and is only modestly larger than the response exactly at ω_0 (≈ 0.667 m). With light damping the peak is sharp and close to ω_0 — which is precisely why driving a lightly damped structure near its natural frequency is dangerous.